INCH-POUND
MIL-PRF-3098/102B
1 October 1997
SUPERSEDING
MIL-C-3098/102A (USAF)
14 March 1975

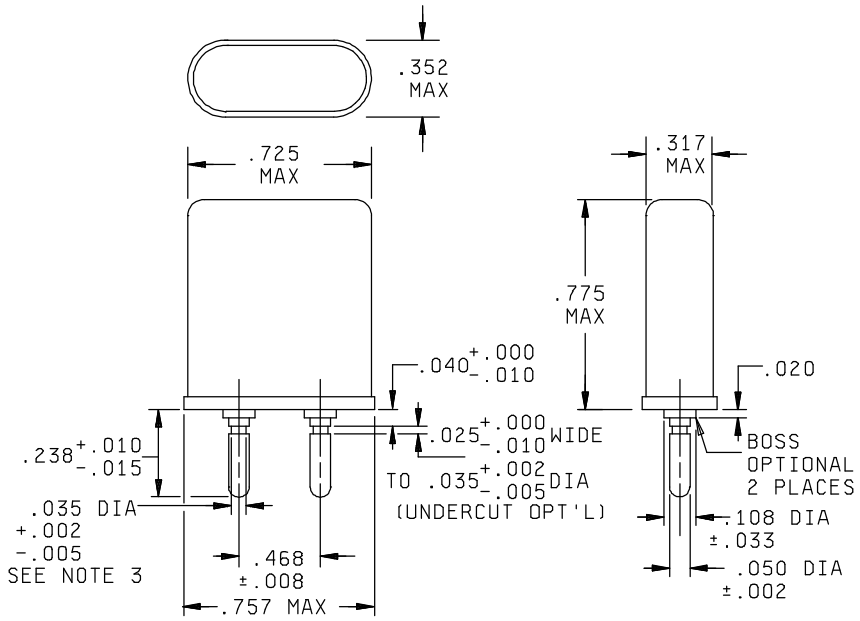
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR125/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 1.85 MHz to 3.05 MHz; fundamental mode; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.020	.51	.050	1.27	.486	12.34
.005	.13	.025	.64	.108	2.74	.725	18.42
.008	.20	.033	.84	.238	6.05	.757	19.23
.010	.25	.035	.89	.317	8.05	.775	19.69
.015	.38	.040	1.02	.352	8.94		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR125/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 1.85 MHz to 3.05 MHz, inclusive.

Capacitance, shunt: 7 pF, maximum.

Frequency tolerance: ±0.005 percent.

Mode of oscillation: Fundamental.

Antiresonance, load capacitance: 32.0 pF ±.05 pF.

Rated drive level: 1.0 mW, maximum.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Equivalent resistance: See table I.

TABLE I. Equivalent resistance.

Frequency Range Inclusive	Maximum Resistance
MHz	Ohms
1.85 to 1.999999	550
2.0 to 2.249999	500
2.25 to 2.999999	320
3.0 to 3.050000	175

Shock:	Below 2 MHz	2 MHz and above
Frequency change permitted:	±0.001 percent	±0.0005 percent

Equivalent resistance change permitted: ±15 percent or ±2 ohms, whichever if greater.

Vibration: Method 204, MIL-STD-202, test condition B; amplitude to produce 5 gravity units, maximum.

	Below 2 MHz	2 MHz and above
Frequency change permitted:	±0.001 percent	±0.0005 percent

Equivalent resistance change permitted: ±15 percent or ±2 ohms, whichever if greater.

Temperature cycling: ±0.001 percent.

Aging:

Frequency change permitted: ±0.002 percent.

Equivalent resistance: See table I.

Aging temperature: 105°C for 200 days.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army - CR
Navy - EC
Air Force -85

Review activities:
Air Force - 11, 17, 19, 99

Preparing activity:
DLA - CC

(Project 5955-0697-66)

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